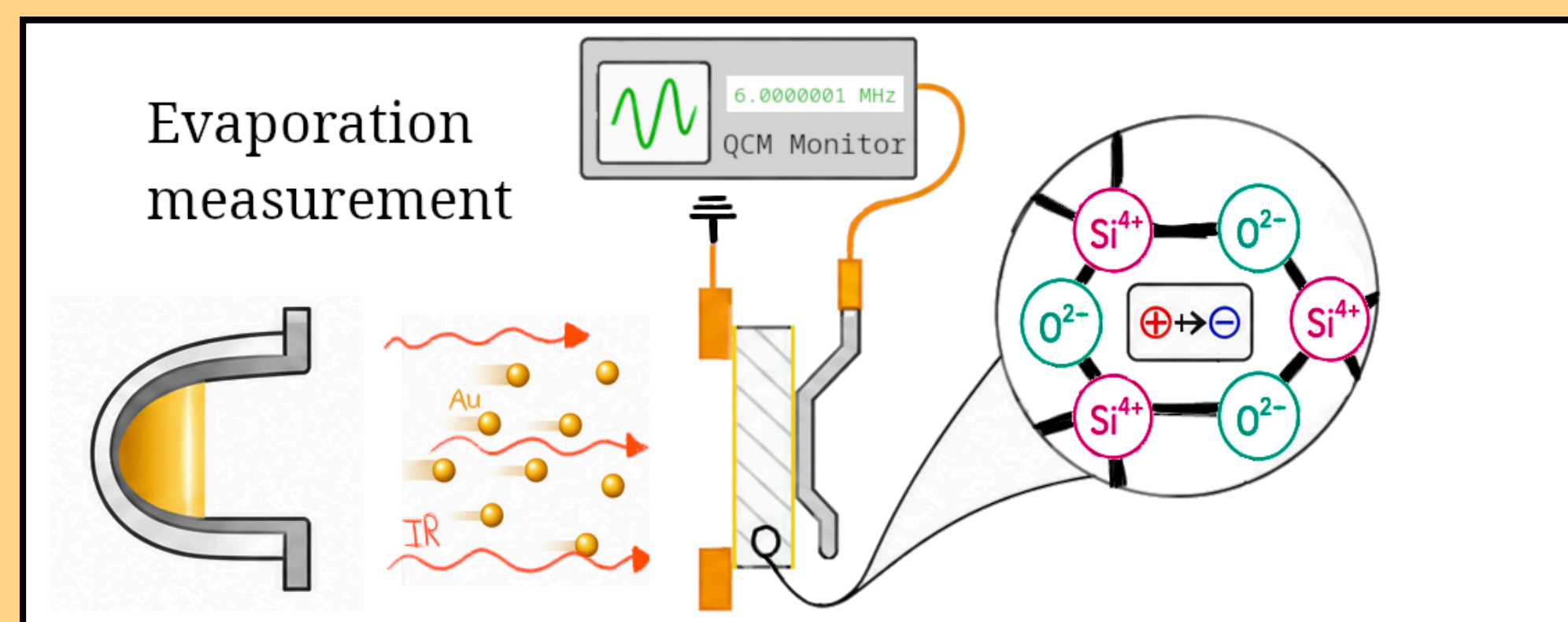


HIGH-PRECISION MULTI-METAL DEPOSITION WITH ACTIVE-COOLED ROTATABLE QCM

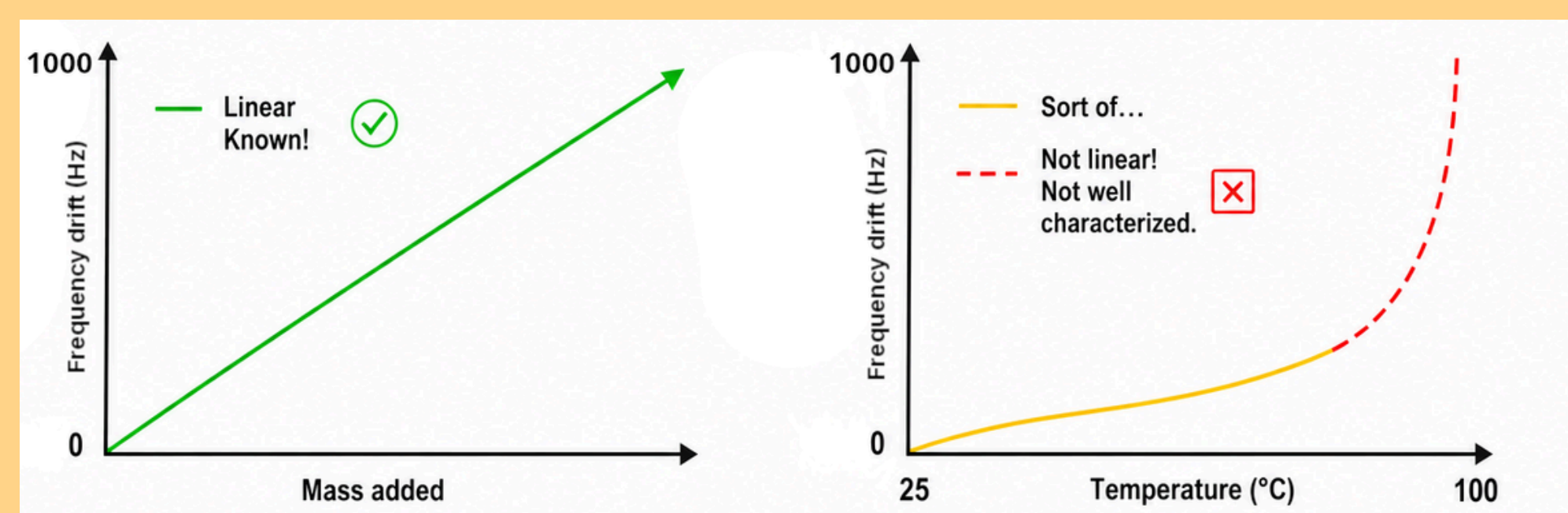
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01. INTRODUCTION

Thin metallic films are a large part of the semiconductor fabrication process, forming interconnects typically made out of precious metals like gold or silver. Thickness control is necessary for quality and reducing metal costs. Usually thermal evaporation under ultra high vacuum is used, where atoms of heated metal will evaporate and form a “beam” without air interference.



This exposes the substrate to a large amount of radiated heat, due to the necessary line of sight. The deposition monitor is a piezoelectric oscillator that detects mass through a change in frequency. Heating relaxes the crystal, allowing it to vibrate slower without extra mass, causing inaccuracy in measurement.



Solutions exist for stationary crystals, but due to UHV sealing direct water piping to a movable crystal isn't possible. Software correction also becomes extremely difficult since the crystal error is not linearly dependent on temperature. Our sponsor requires a rotatable crystal with high accuracy, and so we have developed a hybrid solid-liquid cooling solution, along with a test chamber and validation scheme. There do not appear to be attempts to build cooling for this specific situation in literature.

02. INVESTIGATION AND METHODS

1. How can we know the temperature at the crystal without interfering with its oscillation?

- We measured the temperature on an inactive crystal and two other points in our stage and cooling loop, and built a model to infer crystal temperature from the other two points at steady state, as well as dissipation wattage.
- To test heating without evaporation, a tungsten filament was used
- All of this must be done in a custom UHV chamber!

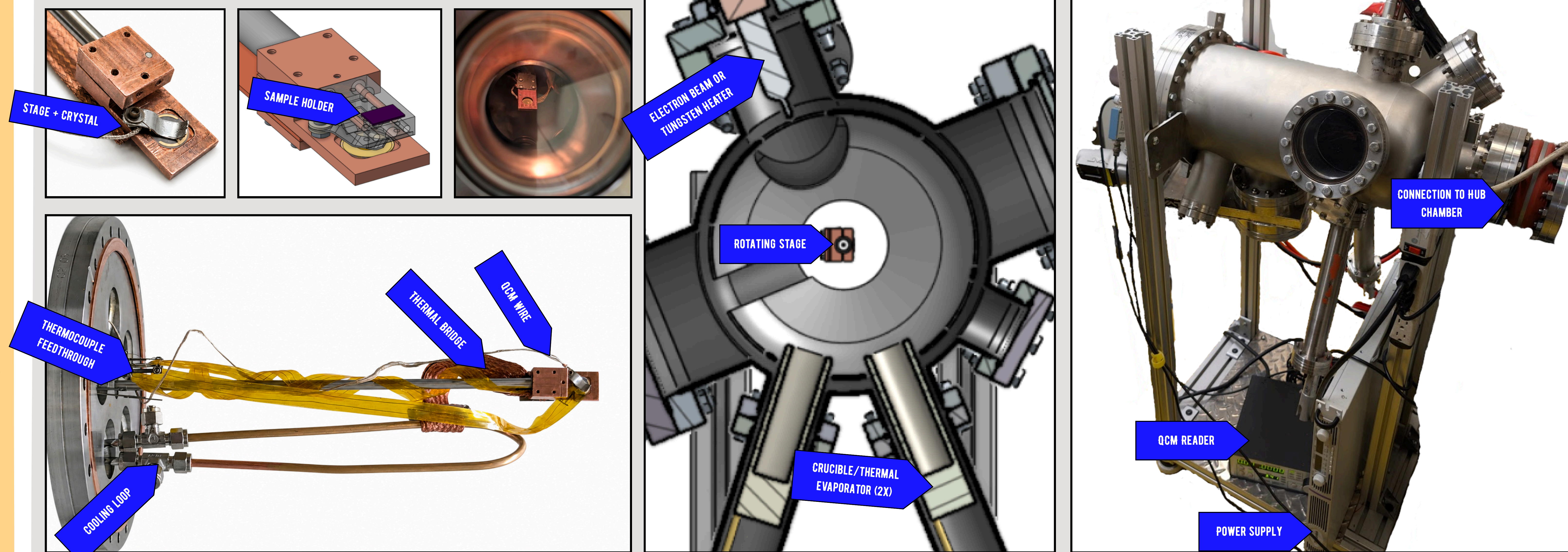
2. How much frequency drift do we observe due to temperature in the uncooled system? How much does cooling actually help?

- We will activate the crystal monitoring and record inferred temperature vs. frequency shift again using tungsten heating

3. Can we deposit consistent and accurate layers?

- Right now we are testing with aluminum evaporation as a cheap alternative to gold
- We plan to test the height by measuring the difference between a deposited and masked region with an atomic force microscope

03. HARDWARE / APPARATUS



04. RESULTS AND FINDINGS

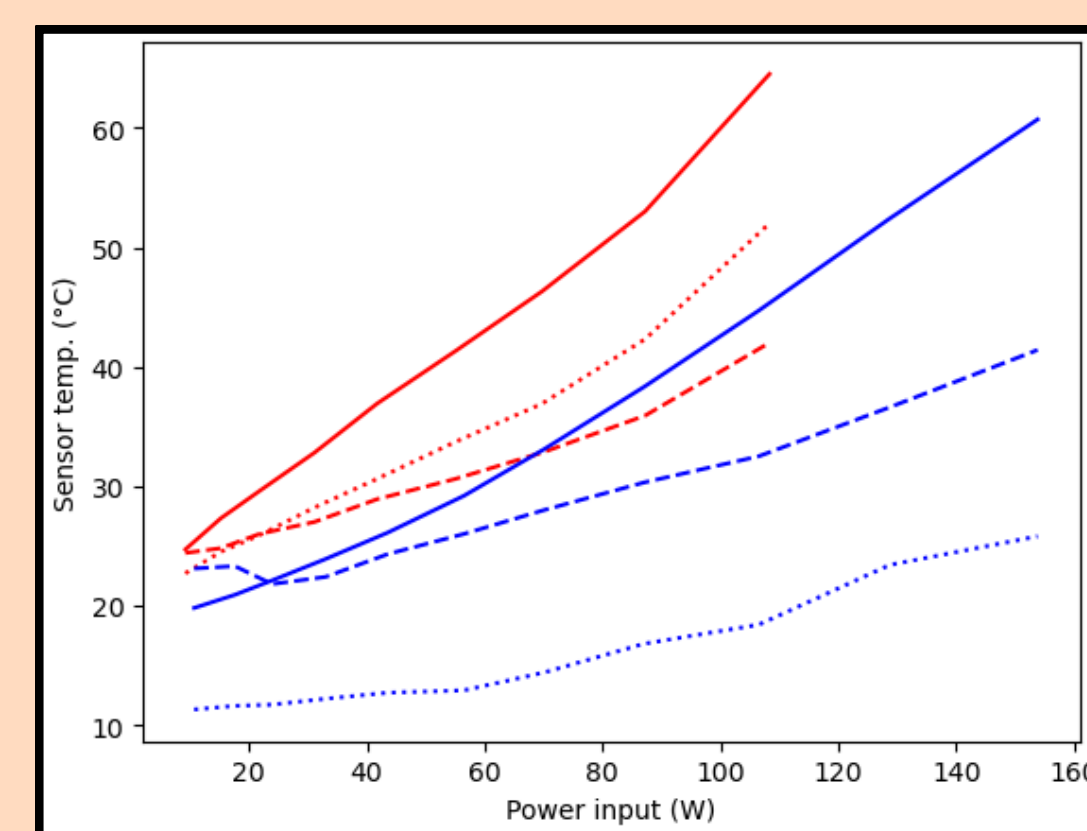


Figure 1:

Input wattage vs. temperature of crystal (solid), off-crystal thermocouple/braid to stage (dashed), and braid to cooling loop joining (dotted). Note the reversal of the latter two for uncooled, and the improvement from cooling. Also note the crossover, on the cooled stage/crystal.

Figures 2,3:

Models correlating off-crystal thermocouple to on-crystal temperature for cooled and uncooled respectively, linearity is due to high conductivity of solid copper stage.

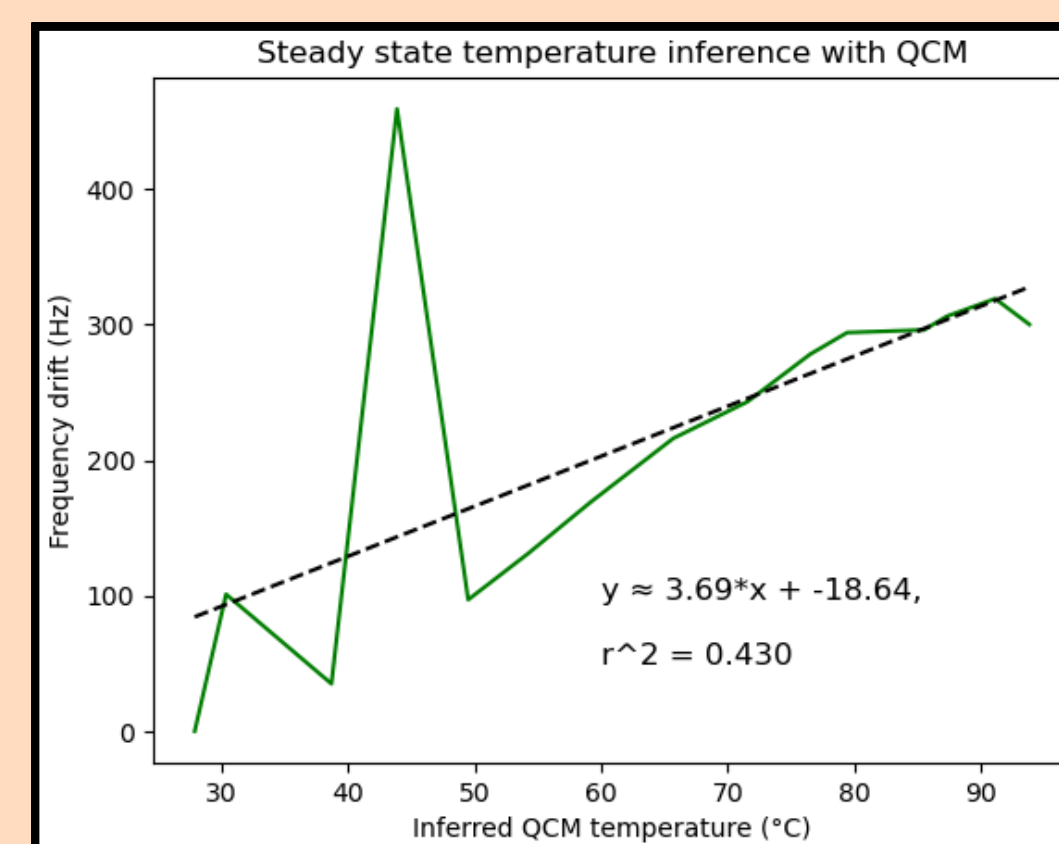
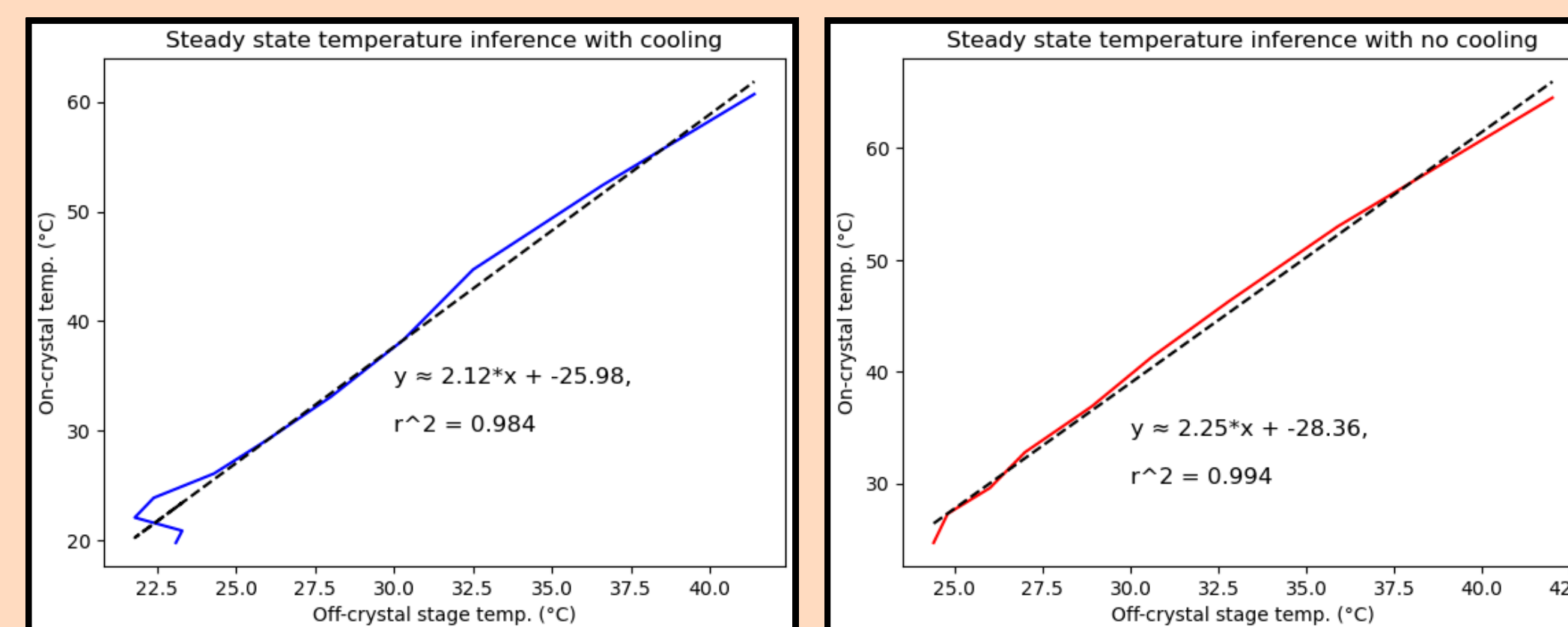


Figure 4:

Inferred QCM temperature vs frequency drift, each 90 Hz is equivalent to about 1nm of gold. This trial was done without cooling due to time constraints and mechanical issues.

Data collection for temperature was performed using LabView and compatible Type K thermocouple readers. For QCM frequency an Inficon monitor and compatible crystal was successfully connected and a single heated trial was run, but we were not able to observe the cubic curve described in literature.

05. CONCLUSIONS AND FUTURE WORK

The cooling system functions well, providing the ability to use double the filament wattage and maintain the same temperature. Improvements could be made to reduce thermal leakage around the heating and deposition apertures, such that the radiant flux will be even more concentrated and the cooling system will be even more effective.

The steady state model based on this system allows prediction of crystal temperature during regular operation, but is limited to literature values for safe temperatures. Further work is needed to better characterize our specific QCM's response to heat and determine the correct limit temperatures for Dr. Bartel's desired accuracy.

Finally, an actual deposition test must be done. The system has been designed such that the sample and QCM are the same distance from the evaporators and both cooled, so no corrections need be made for differences between them. One or two runs should suffice to establish accuracy, and arrangements have been made for the resulting films to be analyzed with an atomic force microscope (AFM) for a precise thickness measurement.

Although we could not validate it fully, the cooling provides enough heat removal to stay below 50°C at 100W (upper bound for regular evaporation flux) and our temperature inference and QCM measurement are functional and ready to collect more accurate data. Overall, a partial success.

06. REFERENCES

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